

Experiment log — 2026-09-04 to 2026-09-18

Every change tried against the volume error, what it was measured against, and what it did. Nothing here is committed to the pipeline; every row is an offline experiment on the run outputs, so the pipeline's own numbers are unchanged throughout. Ground truth is tape unless a row says water.

The one number to keep in mind: on a rigid slim can the pipeline's girth is ****+0.9%**** against calipers. Everything above that on a limb is limb, scale, or wrap – not the model's ability to measure a circumference.

1. Marker detection

change	tried on	result	kept?
White and black cords instead of green	`sunshine\_v2`	Stage 0 (colour-agnostic detector) boxed both bands on 7/7 frames; Stage 3's colour rule could not separate a neutral cord from skin; every plane rejected, no cut	no – cords re-dyed green
Per-band colour instead of one capture average	`sunshine\_v2`	Averaging white + black gave grey RGB[71,64,64]; per-band gave RGB[151,140,142] and [41,34,32]	reverted – moot once cords were green
Keep the projected plane when its colour twin is later gated out	`sunshine\_v2`, cohort 0–6	Recovered the cut on `sunshine\_v2`; on the cohort the same defect had destroyed the lower band on `0\_left`, `2\_left`, `5\_right` (sliver / runaway volumes)	reverted then; ****fixed 2026-09-19****, section 11
Drop foot slices from the limb-axis fit	`sunshine\_v2`	Real ankle band went from 35.7° (rejected at 35°) to 26.1°	reverted with the above

2. Ground truth

change	result
Water displacement (weigh tub before/after)	Carries two biases: film on the withdrawn limb (+20–50 g, 7–17% of an ankle reading) and fill-level shortfall below the overflow point (–100 cm³ per mm on a leg-sized tub). Used on captures 0–6 and `test5`
Circumference tape, disc model, 2 cm spacing	Chosen as primary. Two passes on `test5`: 0.8 cm mean / 2 cm worst on girth, 2 cm on band separation (7% of volume). Water and tape agreed to 4% on `test5` (2070 vs 1985)
Two-point cone through the band girths only	Under-reads the segment by 12–22% (misses the calf bulge). Not usable as truth
Circle-from-circumference bias	Over-reads area by 1.3% at a/b 1.2, 2.6% at 1.3, 6.3% at 1.5. Limbs measured a/b 1.1–1.5. Handbook corrected from “~2%, constant”

3. Where the error is — per capture

Volumes are the span between the two bands unless noted. "wrap" is the alpha fallback's inflation, measured directly on `test6`.

capture	subject	girth err (low / high band)	volume err	alpha fallback	notes
test1	M, hairy	+3.1% / +10.5%	no volume truth	no	four capture styles gave the same error
test2	M	+6.8% / +14.3%	no volume truth	yes, $\chi=-2$	worst residual, 4.08 mm
test3	M	+7.3% / +10.1%	no volume truth	no	cube 10.8 × 10.9 × 10.7 – best cube
test4	M	+5.3% / +10.9%	no volume truth	no	100% ring coverage, still +10.9%
test5	M, hairy shin	+1.5% / +6.7%	**+24.9%** (water 2070)	yes, $\chi=-4$	profile: shin +19%, calf belly +4%
test6	nearly hairless	+7.9% / +13.2%	**+38.7%** (tape 1399)	yes, $\chi=-22$	profile flat +8–12%, no shin spike
fanta_orange	rigid can	**+0.9%** (calipers 18.5)	+5–10% (height +6.8%)	yes, $\chi=-12$	girth exact; height is end caps
fanta_red	can, mirror floor	–	15,164 cm ³	yes	floor removal took 19.5%; cube fill 0.359

`test6` decomposed exactly:

cube-derived scale	+4.2% linear	×1.133	ruler 27.5 vs pipeline 28.7 between bands
residual girth	+5.5%	×1.114	profile vs tape, scale removed
alpha wrap		×1.10	mesh section / points, summed over slices = 1.120
product		×1.387	measured +38.7%

4. Hypotheses tested and ruled out

hypothesis	test	verdict
Upper cut plane sits too high	green band located from mesh colour vs plane height	plane within 0.05–0.7 cm of the band on every sound capture. Wrong – my own earlier claim
Girth over-read comes from incomplete ring coverage	coverage vs error across captures	test4 at 100% coverage still +10.9%. Wrong – my own earlier claim
Leg shake during capture	error vs height (–0.32) and vs surface scatter (–0.19)	four capture styles agree anti-correlated with both. Ruled out
Stray outer points inflate the fitted ring	six trim rules at the band slices	shell is 15 mm thick with an inward tail; best rule moved girth 0.4%. Ruled out
Reconstruction method inflates girth	Poisson vs alpha_shape on test1	circumference byte-identical (it is read from the Stage 3 cloud); alpha volume +17% worse. Ruled out for girth
Vertical blur of the profile	Gaussian blur + gain fit to the tape profile	rms 1.78 → 1.52 cm; systematic +2 cm left on the shin. Weak, not the cause
Cube is physically ≠ 10 cm	can girth +0.9% on a scene whose cube read 0.815 fill	would have read +4% if the cube were 9.6. Probably per-scene reconstruction, not the object – caliper still pending
Hair	test5 (hairy) vs test6 (nearly hairless) profiles	shin spike +19% present only on test5. Supported

5. The wrap, seen directly

`ring\_gallery.png`: `test6` sliced every 2 cm with the watertight mesh's cross-section drawn over the cleaned points.

h cm girth mesh area / points' ellipse
0-12 20-27 1.03 - 1.08
14-28 29-32 1.13 - 1.17 one side carries a second arc ~1 cm inside the
 outer one; the 30x alpha bridges the outer strays
summed over the span: 1.120 (+12.0% volume from the wrap alone)

Poisson tunnelled (χ from -2 to -22) on 5 of 8 runs, which is what forces the alpha fallback. A finer alpha search between the ladder's rungs found nothing tighter than the 25x it chose on `test5`.

6. MLS radius — the one change that moved the volume

Replays Stage 3's ghost chain from the pre-MLS cluster with only the MLS radius varied; then re-wraps. The can is the control.

CAN (girth 18.5, cylinder ≤ 394.9)				TEST6 (tape 1398.6)			
mls	girth	poisson χ	volume	girth	poisson χ	volume	
0	+0.7%	-6	—	+10.5%	-30	—	(alpha 2141)
4	+0.5%	-12	—	+10.0%	-6	—	(alpha 1982) <- pipeline today
8	+0.6%	-2	—	+9.3%	0	—	(alpha 1780)
12	+0.6%	2	384.2	+9.2%	2	1606.6	+14.9%
16	+0.6%	2	385.5	+9.3%	2	1615.7	+15.5%
24	+1.5%	2	392.0	+0.6%	-2	—	<- artefact, see below
32	+3.7%	2	407.9	-12.7%	-44	—	

****12-16x spacing closes Poisson on both objects**** — no fallback, no wrap — and takes `test6` from +38.7% to +15% with the can within 3% of its bound.
****Girth does not move**** at any radius up to 16x, which is what a merged double sheet should do: the merged ring lands between the two sheets, where the ellipse fit already was. The +15% that remains is cube scale and the real over-read, now visible instead of buried under the wrap.

The 24x row on `test6` reads +0.6% and looks like a fix. It is not: the can at the same radius moves the *other* way (+1.5%, then +3.7% at 32x) while `test6` collapses to -12.7%. Opposite signs on the two objects is distortion crossing zero by accident.

****Stacked passes**** (the pipeline's 4x first, then a wide pass):

CAN				TEST6			
mls	girth	poisson χ	volume	girth	poisson χ	volume	
4-12	+0.6%	-2	—	+9.7%	-4	—	
4-16	+0.6%	2	384.4	+9.6%	2	1614.5	+15.4%
4-24	+0.6%	2	386.4	+10.0%	2	1632.1	+16.7%

Same volume as the single 12-16x pass, same untouched girth – and more robust: `4→24` holds the can at +0.6% and `test6` at +10.0% where the single 24x pass distorted both. Denoising each sheet first evidently keeps the wide pass from tearing the surface. ****4→16` is the candidate\*\***: Poisson closes on both objects, the can lands at 384 against a 395 bound, `test6` goes from +38.7% to +15.4%, and nothing about the girth is hidden.

What this change does and does not do, stated plainly: it removes the alpha fallback and its wrap inflation (the $\times 1.10$ term). It does not touch the cube scale ($\times 1.133$) or the residual girth ($\times 1.114$), which together are the +15% that remains. Those need the caliper and a different fix respectively.

****Cold run through the real pipeline**** (wired in behind `MLS\_SECOND\_RADIUS\_MULT`, default off, run with it set to 16; full detail in `COLD\_RUN\_REPORT.md`):

		today (4x)	cold run (4→16)	truth
test6	volume	1939.4 +38.7%	1703.9 +21.8%	1398.6 tape
	recon	alpha 30x	Poisson $\chi=2$	
	mesh/pts	1.105	0.989	
	girth	20.14 / 32.72	19.95 / 32.09	19.0 / 28.0
can	volume	415.5	387.2	≤ 394.9
	height	15.48 +6.8%	14.78 +1.9%	14.5
	girth	18.45	18.45	18.5

Poisson closed in the pipeline's own path on both objects; the cube was byte-identical between runs; the offline sweep's per-slice sections matched the real run to 0.001 but its volume was 5% low. Girth moved about 1% inward.

7. Environment failures found along the way

failure	symptom	cause	fix
Reflective floor	`fanta\_red` 15 L "can", cube fill 0.359	RANSAC removed 19.5% of the floor instead of ~70%; reflections form a second sheet	shoot on the matte tile
WSL GPU degraded	inference 14 s → 320–520 s, GPU 100% / 7.9 GB with nothing else on it	not memory, not swap (both checked and wrong at first)	`wsl --shutdown`; it also recovered on its own overnight
Background job from another repo	~2.7 GB RAM held by a headless Blender render	`SCG/auto\_entity\_extraction` runner, state `T` (suspended)	killed; unrelated to this project
Stage 0 refuses a frame	`test5` img7 "cube missing"	cube occluded behind the ankle in that view	correct rejection; ran on the 7 clean frames

8. The cube scale, checked against the 5 cm markers — 2026-09-19

The cube is 10.0 cm and its printed marker 5.0 cm (measured; the cube was 3D printed). ``pipeline/stages/prep.py`` still carried the cardboard cube's ``FACE_CM = 14.0, MARKER_CM = 6.3``; corrected to 10.0 / 5.0. Only their ratio is used, to map a marker's corners out to its face for the Stage 0 crop window and clipping gate, so the volume never saw those numbers.

The scale itself was then checked a second way (``marker_scale_check.py``): on every frame VGGT was given, detect the markers, read the predicted 3D point under each corner, and measure the marker's edges in scene units. 5.0 cm over that edge is a cm-per-unit that uses no mesh at all.

	marker edge at cube scale		marker scale / cube scale
	pointmap	depth	linear (pointmap / depth)
test1	4.85	4.78	1.030 / 1.046
test2	4.89	4.86	1.023 / 1.028
test3	4.86	4.84	1.030 / 1.033
test4	4.95	4.87	1.010 / 1.026
test5	4.86	4.82	1.029 / 1.038
test6	4.89	4.88	1.023 / 1.024
fanta_orange (can)	4.96	4.98	1.008 / 1.005
	true 5.00		

The cube-volume scale is not inflated. It reads the marker 1.5–3% ***small*** on the limbs and under 1% small on the can, and the marker's vertical and horizontal edges agree (``test6`` 4.91 / 4.92, ``test5`` 4.85 / 4.86), so nothing is stretched along one axis. The cube mesh's extents at cube scale are 11.5 × 11.6 × 10.9 cm: the mesh bloats sideways from the ghost sheets on its faces while its volume stays near 1000, which is why a volume-derived scale is the more robust of the two and a face-distance scale would be 15% worse.

This retracts the 2026-09-18 decomposition's "cube-derived scale +4.2% linear" term. Under the marker scale the numbers get slightly **larger**, not smaller:

		cube scale	marker scale (pointmap)	truth
test6	band separation	28.66 cm	29.3 cm	27.5 ruler
test6	volume (cold run)	1703.9	~1805	1398.6 tape
can	girth	18.45	18.6	18.5
can	height	14.78	14.9	14.5

So with the cube at 10.0 and the marker at 5.0 both confirmed, VGGT's own geometry puts the bands 4–6% further apart than the ruler, and the girth over-read is the whole remaining error, not a scale term plus a girth term.

One pattern fits all seven captures: the girth error grows with height above the floor cube – `test6` +5% at the ankle band, +15% at the knee band, and the same rising shape on `test1`–`test5` – while the rigid can, which stands on the floor next to the cube and is only 14.5 cm tall, reads within 1–3%. The reference sits at the floor; the measurement is 20–30 cm above it. Scale drift with distance from the reference is the open hypothesis. It is testable with a second known length up at band height (a ruler standing beside the leg, or the cube raised to the upper band) – a capture, not a code change.

****Field of view, checked against the phone's own lens.**** The originals (`inputs/test6/\*.zip`, iPhone 17 main camera, 26 mm equivalent, 4284 × 5712) give a true focal of 4290 px, which is 518.8 px on the full-width 518 crop VGGT was fed. VGGT's predicted focal, per frame, against that:

scene	VGGT / true focal	volume error
test3	1.01 – 1.06 (median 1.037)	girth +7 / +10%
test5	1.01 – 1.07 (median 1.047)	+24.9%
test6	1.02 – 1.07 (median 1.037)	+21.8% (cold run)
fanta_orange	1.08 – 1.18 (median 1.104)	girth +0.9%, height +1.9%

VGGT's focal runs 2–7% long on the leg scenes (a narrower field of view than the lens has) and 10% long on the can scene, assuming the can was shot with the same lens. The scene with the largest focal error is the one that reads right, so the focal error does not track the volume error. VGGT takes no intrinsics as input, so the true focal cannot be supplied to it; a model that accepts a known field of view would be the way to test this further. Camera geometry is otherwise ordinary: phone ~32 cm above the floor, 6–17° pitch down, cube and upper band at the same distance from the camera (ratio 1.07 on `test6`, 1.01 on `test5`).

9. The original photos, and the bug they exposed — 2026-09-19

`test6` was re-run from the phone's HEIC originals (4284 × 5712, iPhone 17) instead of the 1108 × 1477 LINE copies, converted to JPEG in `inputs/test6\_orig/`. Same settings as the cold run (`MLS\_SECOND\_RADIUS\_MULT=16`).

	volume	err	recon (leg / cube)	voxel	leg pts
LINE copies (cold run)	1703.9	+21.8%	Poisson / Poisson	0.0046	27,244
originals, same code	1848.9	+32.2%	Poisson / Poisson	0.0079	10,012

Worse, and VGGT's geometry had barely moved (marker edge 4.90 both; green band separation 28.95 vs 28.66; raw-cluster girth at the upper band 33.2 vs 32.3). The difference was in Stage 3: the denser cloud (762k vs 347k points after the 2 mm voxel) got a ****coarser**** ghost voxel, and every radius downstream (dedup, normal filter, MLS 4× and 16×) hangs off that voxel.

****Root cause**** (`pipeline/ghost.py`, `compute\_voxel\_size`): the "mean nearest-neighbour distance" was measured on a random 5,000-point sample **against itself**. That is the sample's spacing, which grows with the square root of the cloud's size, not the cloud's:

	real spacing	what the code measured	voxel (x0.65)
LINE copies	0.0016	0.0071	0.0046
originals	0.0019	0.0123	0.0079

Fix: build the tree on every point, sample only the queries. `GHOST\_VOXEL\_FACTOR` went from 0.65 to 2.8 so the validated runs keep the voxel they had: `test6` LINE gives 0.0046 exactly as before and reproduces (1704.0 vs 1703.9, Poisson $\chi=2$ both, girth 19.95 both). The originals now get 0.0054 instead of 0.0079. Every constant in the ghost/MLS chain was tuned against the inflated value, which is why the factor is rescaled rather than left at 0.65.

****Originals with the fix****: 1625.0 (+16.2%), leg Poisson $\chi=2$ with 23k points like the LINE run – but the ****cube**** fell to the alpha fallback (Poisson $\chi=-2$) and came out $12.5 \times 12.2 \times 10.2$ cm, so the volume-derived scale is 6% small (marker edge reads 4.72 at cube scale). That number is a scale artefact, not an improvement. At the marker scale, which does not depend on the cube mesh, the three `test6` runs line up:

	cube scale	marker/cube	at marker scale
LINE copies, fix	1704.0	1.027 lin.	~1845
originals, no fix	1848.9	1.020	~1963
originals, fix	1625.0	1.060	~1936
			truth 1398.6

****Can control with the fix****: voxel 0.0040 → 0.0045; the cube now closes under Poisson ($\chi=2$, it had fallen to alpha at $\chi=0$) and measures $10.6 \times 10.6 \times 10.2$ cm; can volume 387.2 → 394.3 against the 394.9 cylinder bound, height 14.78 → 14.92 (truth 14.5). Control held.

Full-resolution input does not help: VGGT returns a leg 3–5% fatter from the originals than from the LINE copies, with the same cube. Input pixels move the answer by a few percent either way; that is the model's noise floor, and the +30% at marker scale is the same girth over-read as before.

10. Second MLS pass validated, now the default — 2026-09-19

Four cold runs through `run.py` with `MLS\_SECOND\_RADIUS\_MULT=16` and the voxel-spacing fix in, against water truth. test5 needed `--continue-on-rejected` (one frame has the cube occluded).

	truth	before (4x only)		now (4x -> 16x)			girth low / high	
test5 water	2070	2585.3	+24.9%	alpha $\chi=-4$	2438.6	+17.8% Poisson	22.16 / 34.55	(tape 22 / 32.5)
0_right water	1830	2485.2	+35.8%	Poisson	2503.9	+36.8% Poisson	19.98 / 33.30	(20.5 / 29.5)
1_left water	1600	2125.8	+32.9%	Poisson	2154.7	+34.7% Poisson	20.80 / 33.95	(21.5 / 32.0)
6_left water	2800	3489.7	+24.6%	Poisson	3507.7	+25.3% Poisson	23.69 / 39.10	(24.3 / 35.0)

Where the fallback used to fire (test5, and test6 and the can in the cold run) Poisson now closes and the wrap's inflation goes; where it already closed the pass moves the volume under 1% and the girth under 0.3 cm. The cube is untouched by it. `MLS\_SECOND\_RADIUS\_MULT` now defaults to 16. Still owed: a cap in centimetres on the second radius, so a sparser cloud does not get a physically larger one.

11. The projected-plane merge defect, fixed — 2026-09-19

`\_merge\_projected\_planes` kept the colour fit whenever Stage 0's projected plane agreed with it, and threw the projected plane away. The gates run later, in levelled space, and the height gate exempts **projected** planes (a band the detector saw on most photographs is its own corroboration) but not colour fits. So on a capture whose ankle band sits below one cube height, the colour fit was rejected, the projected plane that would have survived was already gone, and the band vanished. With one band left the span cut ran between the knee band and whatever third plane existed.

Fix: the projected twin rides along as the colour plane's backup. If the colour fit fails the height gate or the axis gate, the backup takes its place (for the axis gate, only if it passes the same test). Backups are stripped before the planes are published.

	before	after
0_left (water 1800)	ankle colour fit rejected, 18% of height; cut between knee and a third plane: 10.1 cm ³ sliver	colour fit rejected, replaced by the 1,003-pt projected plane; cut ankle to knee: 1773.3 cm ³ (-1.5%)
test6 (regression)	1703.998	1703.996, planes byte-identical

The other two captures the defect destroyed, re-run with the fix:

	before (2026-09-04)	after
2_left (water 1050)	3381.0 +222% band lost	1442.0 +37% both bands; both colour fits failed the axis gate (58°, 35°) and the projected planes took over; girth 19.05 / 30.61 against 18.5 / 27.5
5_right (water 1770)	9337.3 +428% band lost	8437.8 +377% both bands found, but the capture is a mirror-floor failure: RANSAC took 89% of the cloud as floor, 4.5k limb points survive, ring coverage 28-46%, Poisson balloons. Not a planes problem.

The ankle plane after the fix is the projected one (10.3° from vertical, girth 18.90 against a 20.5 tape). The knee plane is still the colour fit, 144 points with a normal 51° from vertical, which the 35° limb-axis gate

lets through because the limb itself leans; its oblique slice reads 40.8 cm against a 30.0 tape. That is a separate defect, now on the list.

12. TSDF fusion as Stage 2, and the inner-sheet idea — 2026-09-19

Full write-up in `TSDF\_STAGE2.md`. Fusing the eight depth maps into one signed-distance surface (`POINTCLOUD\_METHOD=tsdf`, off by default) gives a cube that fills 0.86 of its box instead of 0.73, a scale within 1% of the markers, and Poisson closing with no merge pass. The limb does not shrink: girth at the bands identical to the default path, volume 1686 in an offline replay and 1778 through `run.py` against 1704 today. The can control failed on clustering both times. Three other attempts on the same predictions: picking the inner of the two ghost sheets (confidence does not separate them; the inner mode sits only 1–2 mm inside the fit and leaves the knee at +12%), VGGT at 1022 px (breaks: cube fill 0.37), and a perfect cube fitted to the mesh's faces for scale (6.5% off the marker scale; it would cancel the leg's error with a manufactured one).

13. Still open

- ~~Caliper the cube edge.~~ Done 2026-09-19: cube 10.0, marker 5.0. The scale is not the error (section 8).
- **A known length at band height.** Ruler beside the leg or the cube raised to the upper band, to test scale drift with distance from the reference.
- **Re-shoot `fanta\_red` on the matte tile** with a ruler on the band separation – the span-on-a-cylinder test.
- ~~The projected-plane merge defect~~ fixed 2026-09-19 (section 11).
- **Colour-fit plane normals** can be far off the limb (51° on `0\_left`'s knee band, 144 pts). The cut still lands on the band but slices it obliquely.
- **`1\_left` drifted 1.2% between identical runs** on 2026-09-04; `test6` reproduced to the byte on 2026-09-18. Not understood.